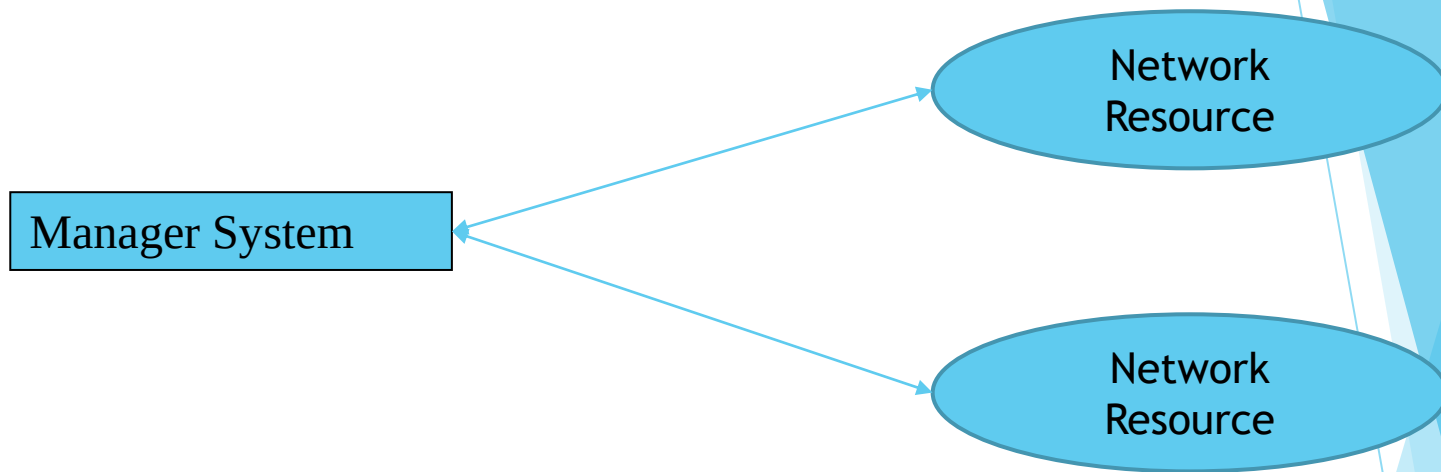


Basic Management Concepts and Abstract Syntax Notation One (ASN.1)

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Management Concepts (Technical View) (1)

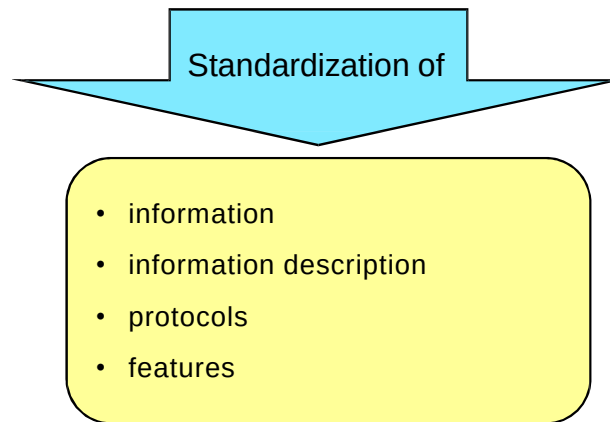


- ▶ Physical vs. logical resources
- ▶ Resources at the level of networks / systems / applications
- ▶ Management Protocols
 - ▶ Inband management
 - ▶ Outband management

Problem areas

- Identification and description of management-relevant information
- Access to the management information
- Further processing / evaluation of management information

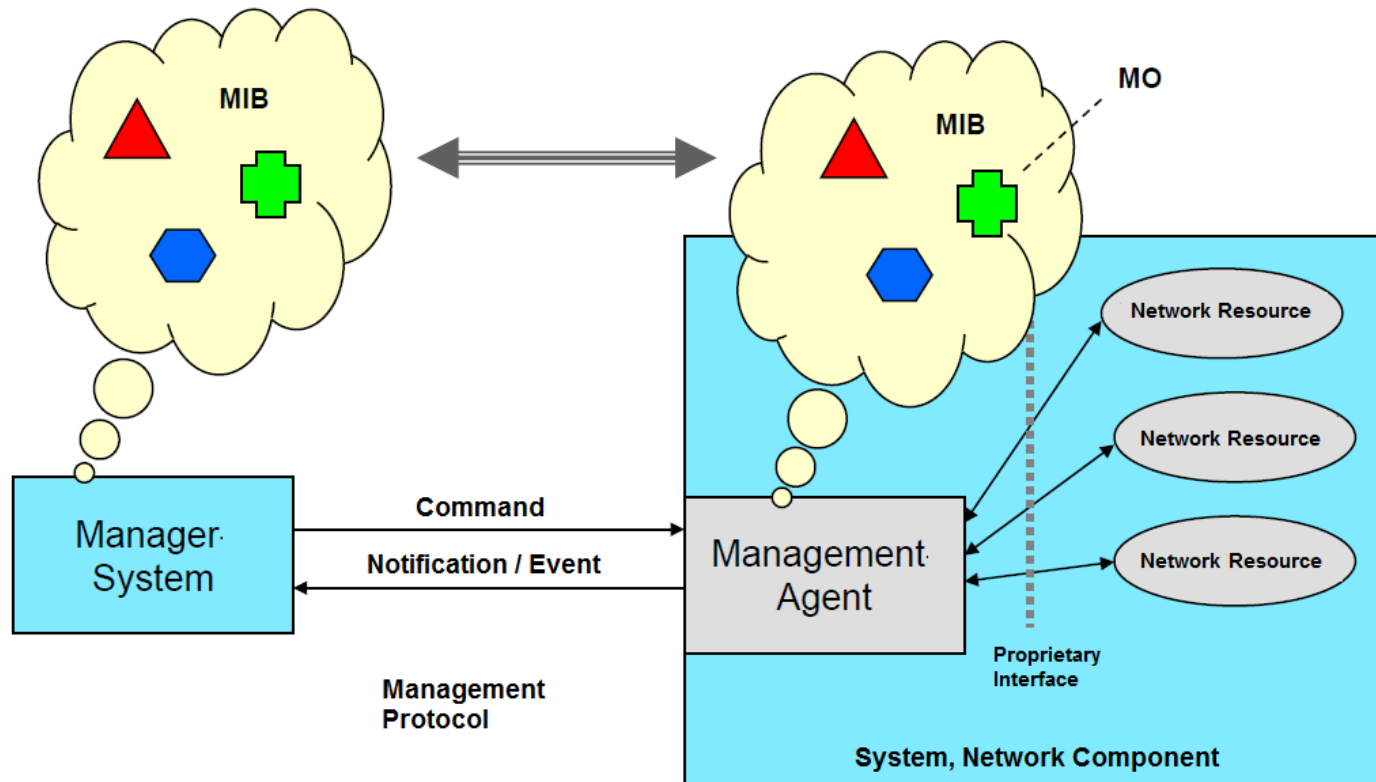
Heterogeneity of networks



Based on this:

- Management tools, management architectures and platforms
- Integration of network management in the corporate context

Management Concepts (Technical View) (2)



Important terms

- **System**, also known as Managed Node or Managed Network Entity
 - = standalone physical entity ("device"), e.g. PC, network component, printer, coffee machine
- **(Network)Resource**
- **Managed Object (MO)** = Management view of a (network) resource
 - MOs are often modeled object-oriented
 - MO element of a managed object class, inheritance, functions
 - Existence of the resource is independent of the existence of the MOs
 - The relationship between MO and resource is implementation-dependent
- **Management Agent**
 - = Management application that provides management information and functionality to a system through a management protocol
- **Manager System**, also known as Network Management System (NMS)
- **MIB** (Management Information Base)
 - = (Partial) amount of management-relevant information in a system
 - MIB is not a database of MOs
 - The MO descriptions are not part of the MIB

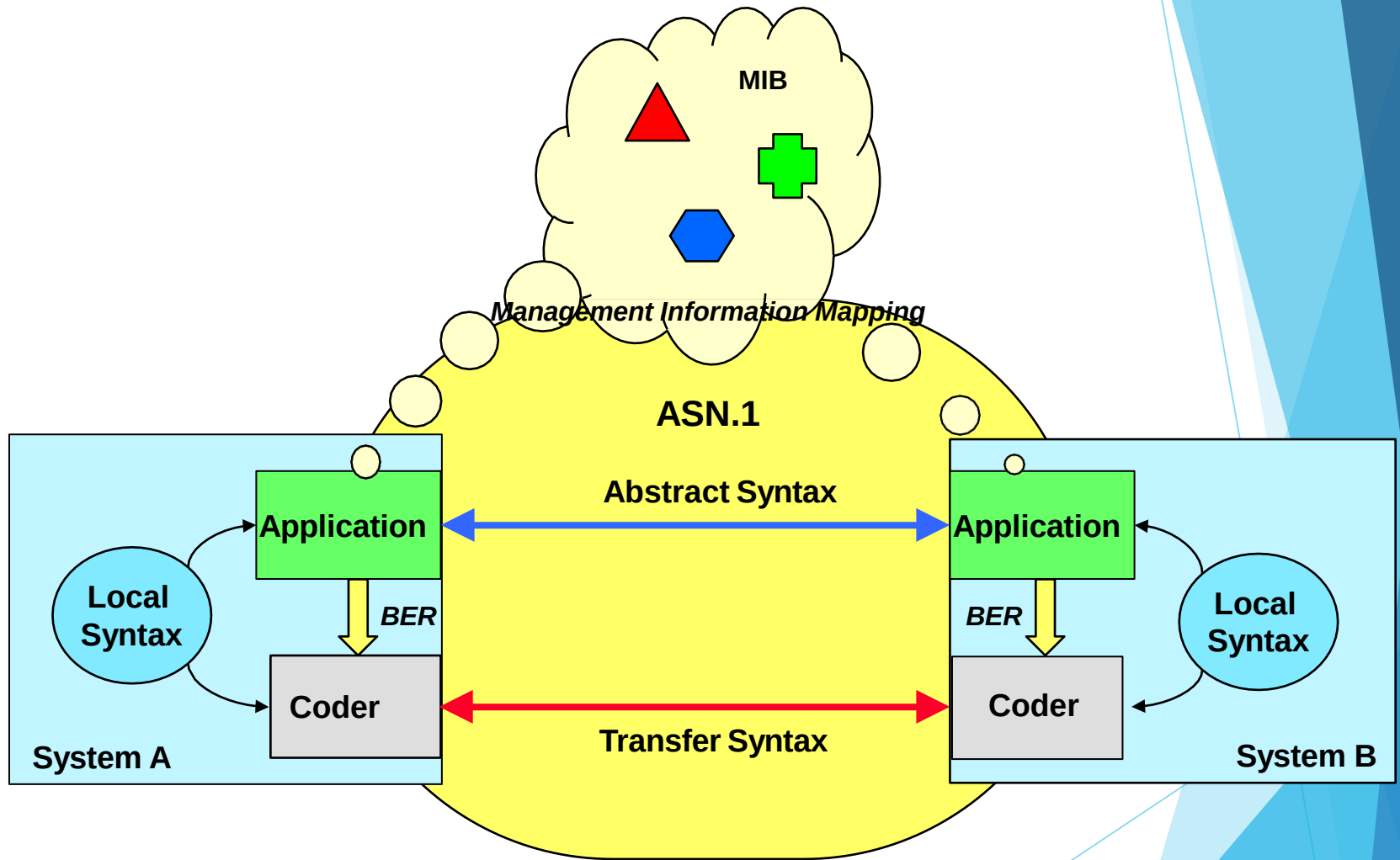
Management-Frameworks

Two main directions:

- Management framework of ISO: OSI management
 - OSI management is the basis of TMN (telecommunications management framework)
 - TMN is the framework for distributed management in the field of telecommunications
 - Management of "big" components, complex
 - real object-oriented
- Management framework of the Internet World: SNMP
 - Pragmatic approach
 - Orientation to LANs and the Internet
 - management of "small" components, easy
 - pseudo-object oriented

Abstract Syntax Notation One (ASN.1)

Abstract Syntax Notation 1 (ASN.1)



ASN.1 – Standardization and use

Standardization:

- ASN.1 - Abstract Syntax Notation One: ITU-T X.680 / ISO 8824-1 (1997)
- BER - Basic Encoding Rules of ASN.1: ITU-T X.690 / ISO 8825-1 (1997)

Applications:

- X.400 Message Handling System / X.500 Directory Services
- Secure Electronic Transaction (SET)
- Telecommunications Management Network (TMN)
- Simple Network Transfer Protocol (SNMP)
 - Definition of the Structure of Management Information (SMI), the MIB and MOs
 - Syntactic structure of SNMP PDUs

ASN.1 Basic Structures

- ▶ ASN.1 Type Definition

- ▶ `<newType> ::= <typeSpecification>`

- ▶ ASN.1 Value Definition

- ▶ `<newValue><type> ::= <valueSpecification>`

- ▶ Comments

- ▶ `--this is a comment`

ASN.1 Simple Data Types

```
<simpleType> ::=  INTEGER |  OCTET  
                  STRING |  
                  OBJECT IDENTIFIER |  
                  NULL
```

Examples of values:

INTEGER: 2, 3, '1e3'h, -34

OCTET STRING: "hallo", 'e1'h

OBJECT IDENTIFIER:

 {1 3 6 1}, {iso org(3) dod(6) internet(1) mgmt(2) mib(1)}

NULL:

Examples of ASN.1 type definitions

```
Counter ::= INTEGER
```

```
IpAddress ::= OCTET STRING
```

```
Months ::= ENUMERATED {january (1),  
february (2),  
march (3),  
april (4),  
may (5),  
june (6),  
july (7),  
august (8),  
september (9),  
october (10),  
november (11),  
december (12)}
```

ASN.1 type restrictions

`<subtype name> ::= <type> ( <constraint> )`

Beispiele:

`Counter ::= INTEGER ( 0..4294967295 )`

`IpAddress ::= OCTET STRING ( SIZE(4) )`

`Spring ::= Months ( march | april | may )`

`Summer ::= Months ( june | july | august )`

`SmallPrime ::= INTEGER ( 2 | 3 | 5 | 7 | 11 )`

`ExportKey ::= BIT STRING ( SIZE(40) )`

`ASCII ::= INTEGER ( 0..'ff'h )`

`Boolean ::= INTEGER {true(1), false(0)}`

`Status ::= INTEGER { online(1), offline(2),
test(3) }`

Examples of ASN.1 value assignments

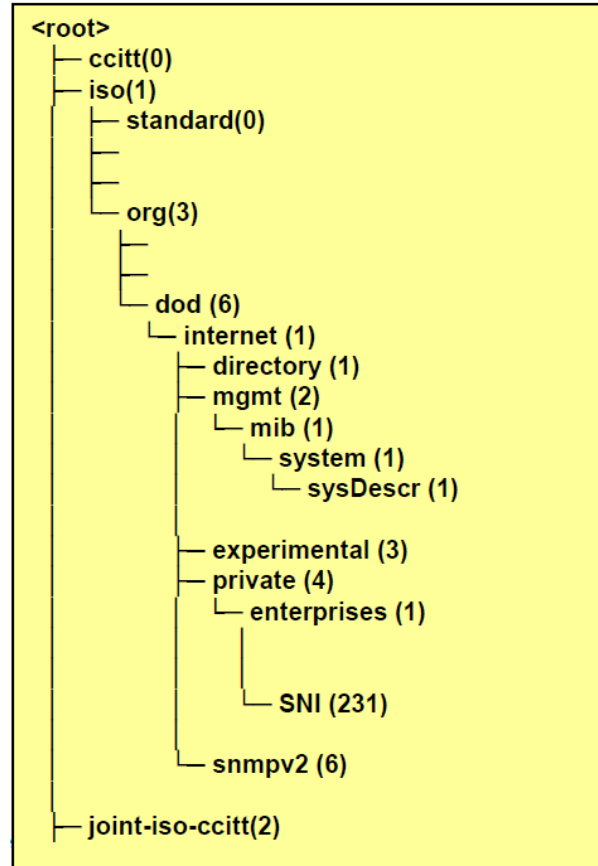
```
ipInReceives Counter      ::= 2450
ipRouteMask IpAddress     ::= 'FFFFFFF0'H
currentMonth Months       ::= february

currentTime UTCTime       ::= "000204075015+0100"
givenName VisibleString  ::= "Martin"
m INTEGER                 ::= maximum(127)  -- maximum as a label
                                   -- the Integer 127 has
                                   -- pure commentary

workstation Status        ::= online(1)
encryptionKey ExportKey   ::= 'A1B2C3D4E5'H
```

ASN.1 Object Identifier Data Type

```
internet OBJECT IDENTIFIER
  ::= { iso(1) org(3)
        dod(6) 1 }
private OBJECT IDENTIFIER
  ::= { internet 4 }
```



Composite types: sequence

- `<seqType> = "SEQUENCE" "{" <seqMember> ["," <seqMember>]... "}"`
- `<seqMember> = lcName <seqDataType>`

Example of type definition:

```
UserAccount ::= SEQUENCE { username OCTET STRING, password
OCTET STRING, accountNr INTEGER
}
```

Example of value assignment:

```
myAccount UserAccount ::= { username "helga", password
"jane57", accountNr 4613
}
```


Composite types: sequence of

- `<seqOfType> = "SEQUENCE OF" <seqTyp>`

Example of type definition:

```
AccountRegistry ::= SEQUENCE OF UserAccount
```

Example of value assignment:

```
myAccountRegistry AccountRegistry ::= {  
    {username "helga", password "jane57", accountNr 4613}  
    {username "otto", password "otto123", accountNr 4614}  
}
```

Type SEQUENCE OF is similar to SEQUENCE, except that all values in the ordered list must be of the same type.

Composite types: choice

Example:

```
CHOICE {case1 INTEGER,  
        case2 OCTET STRING}
```

Tagging

- Objective: To introduce type identifiers for encoding purposes.
- To differentiate: tag-type and tag-value

Type	Tag-Type	Tag-Nr.
INTEGER	UNIVERSAL(0)	2
OCTET STRING	UNIVERSAL(0)	4
OBJECT IDENTIFIER	UNIVERSAL(0)	6
NULL	UNIVERSAL(0)	5
SEQUENCE	UNIVERSAL(0)	16
VisibleString	UNIVERSAL(0)	26
Gauge	APPLICATION(1)	2
TimeTicks	APPLICATION(1)	3

Basic Encoding Rules (BER)

Class:

- ▶ 0 0 = Universal: Defined in ASN.1-Standard
- ▶ 0 1 = Application: Valid for an application
- ▶ 1 0 = Context-specific
- ▶ 1 1 = Private

- Principle: TLV coding

Example: Coding of INTEGER value 5

Tag:

0	0	0	0	0	0	1	0
Class		Structur ed Type (1)	Tag-Number				

Length:

There are two forms: short (for lengths between 0 and 127), and long definite (for lengths between 0 and $2^{1016} - 1$).

0	0	0	0	0	0	0	1
---	---	---	---	---	---	---	---

Value:

0	0	0	0	0	1	0	1
---	---	---	---	---	---	---	---

'020105'h

Tagging: Classes and encoding types

Class:

- 0 0 = Universal: defined in the ASN.1 standard
- 0 1 = Application: valid for one application
- 1 0 = Context-specific: comes from the context
- 1 1 = Private: for your own specifications

Coding types

- Explicit tagging:

```
IpAddress ::= [APPLICATION 0] OCTET STRING (SIZE(4))
```

- Implicit tagging

```
IpAddress ::= [APPLICATION 0] IMPLICIT OCTET STRING (SIZE(4))
```

The length field

Two forms that may occur alternatively:

- Short form:

Bit 8 = 0, bit 7 to 1 = length of the value

- Long form:

Bit 8 = 1, bits 7 to 1 = length of the actual length field (1 to 127)

In the example INTEGER value 5:

Short form:

0	0	0	0	0	0	0	1
---	---	---	---	---	---	---	---

Long form:

1	0	0	0	0	0	0	1
---	---	---	---	---	---	---	---

0	0	0	0	0	0	0	1
---	---	---	---	---	---	---	---

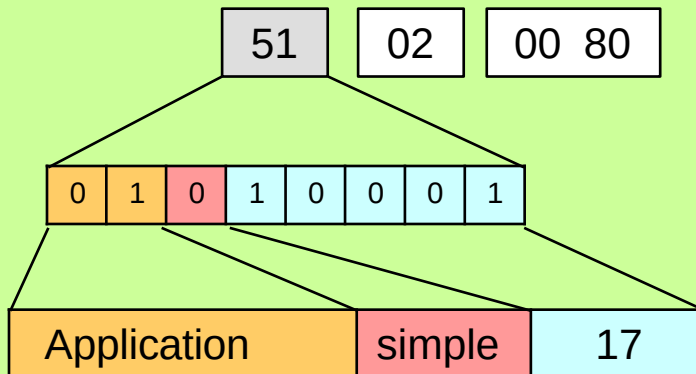
Example

DayOfYear ::=
[APPLICATION 17] IMPLICIT INTEGER

Type Definition

today DayOfYear ::= 128

*Value
Assignment*



BER-Coding

Questions

- How does the agent access the resource?
- Which description technique describes the structure of the MOs?
- How are the MOs clearly identified?
- How are the MO descriptions clearly identified?
- How does manager system and agent communicate with each other?
- Which operations can be performed on a MO (Create?)
- How does the manager system or agent know the MOs or MO descriptions?
- How can the management be structured with several management systems?

Thank you for your attention!